Laboratory 08

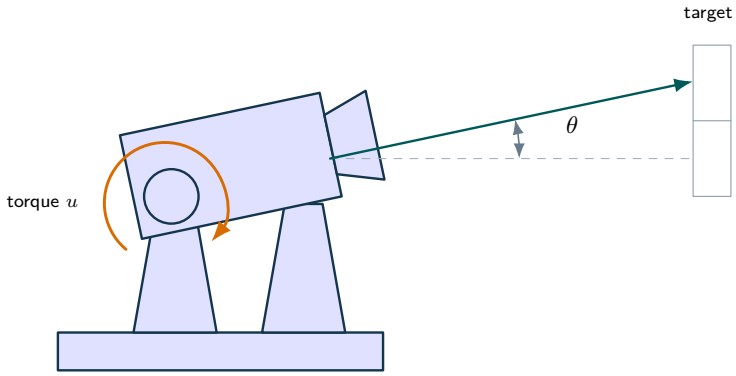
Lead-lag design for a normalized camera axis

Worked solutions – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal camera-axis plant: $G(s) = 1/[s(s+1)]$. Design crossover is 3 rad/s and desired phase margin is 55°. Use unrounded compensator parameters in subsequent calculations.

1. A camera axis with simultaneous speed and phase requirements

The physical sketch shows a single rotating camera axis with a position sensor. The normalized transfer is stipulated; inertia, drive bandwidth and unit scaling are already aggregated into it.



Gimbal axis with angular readout

Normalized model $G(s) = 1/[s(s+1)]$; geometry is illustrative

A camera gimbal illustrates angular tracking and disturbance rejection. Its normalized plant is stipulated, with no claim that geometry alone determines its two poles.

(a) The normalized plant is $G(s) = 1/[s(s+1)]$. At unity controller gain, derive the closed-loop polynomial, gain crossover and phase margin.

Worked reasoning. The closed polynomial is $s^2 + s + 1$, so the baseline loop is stable. Magnitude one requires $\omega^2(1+\omega^2) = 1$, hence $\omega_c = \sqrt{(\sqrt{5}-1)/2}$. The phase is $-90° - \arctan\omega_c$ and the phase margin is $90° - \arctan\omega_c \approx 51.83°$. These values characterize the baseline before a new crossover is imposed.

(b) A redesign targets crossover 3 rad/s and phase margin 55 degrees. Compute the plant phase at that frequency and the phase the lead section must provide. Explain why changing scalar gain alone cannot achieve both targets.

Worked reasoning. At $\omega = 3$, plant phase is $-90° - \arctan 3 \approx -161.565°$. A 55 degree margin requires loop phase $-125°$, so the controller must provide $\phi = \arctan 3 - 35° \approx 36.565°$. A positive scalar gain changes magnitude but not phase at the specified frequency. It can place the crossover at 3, but would leave only about 18.435 degrees of margin.

(c) Suppose an omitted actuator lag $1/(0.1s+1)$ is added to the plant. At the fixed 3 rad/s crossover, find the scalar gain needed for unity magnitude, the resulting phase margin, and the lead phase now required for the 55° target. Compare with the original plant.

Worked reasoning. The augmented magnitude is $1/[3\sqrt{10}\sqrt{1+0.3^2}]$, so the scalar gain is $K = 3\sqrt{10.9} \approx 9.90454$. The lag also subtracts $\tan^{-1}(0.3) \approx 16.69924°$. The remaining margin is $90° - \tan^{-1}(3) - \tan^{-1}(0.3) \approx 1.73570°$, leaving a required lead of approximately $53.26430°$. The original plant needed only $36.56505°$. Restoring crossover by gain adjustment therefore does not recover the original phase budget; omitted actuator dynamics materially change the compensator design.

2. Realizing and normalizing a lead network

The desired phase improvement must be produced by a realizable pole–zero pair and then normalized at the selected frequency.

(a) For $C_{lead} = K(Ts+1)/(\alpha Ts+1)$, center maximum phase lead at 3 rad/s and supply the phase found in Problem 1(b). Derive α and T and locate the controller zero and pole.

Worked reasoning. Use $\sin\phi = (1-\alpha)/(1+\alpha)$, giving $\alpha = (1-\sin\phi)/(1+\sin\phi)$. Centering the phase peak requires $3 = 1/(T\sqrt{\alpha})$, so $T = 1/(3\sqrt{\alpha})$. The zero is $-1/T = -3\sqrt{\alpha}$ and the pole is $-1/(\alpha T) = -3/\sqrt{\alpha}$. Since $0 < \alpha < 1$, the pole lies farther left than the zero. These positions produce positive phase lead between their break frequencies.

(b) Find K so the lead-compensated magnitude equals one at 3 rad/s. Show explicitly why the magnitude of the lead section cannot be ignored.

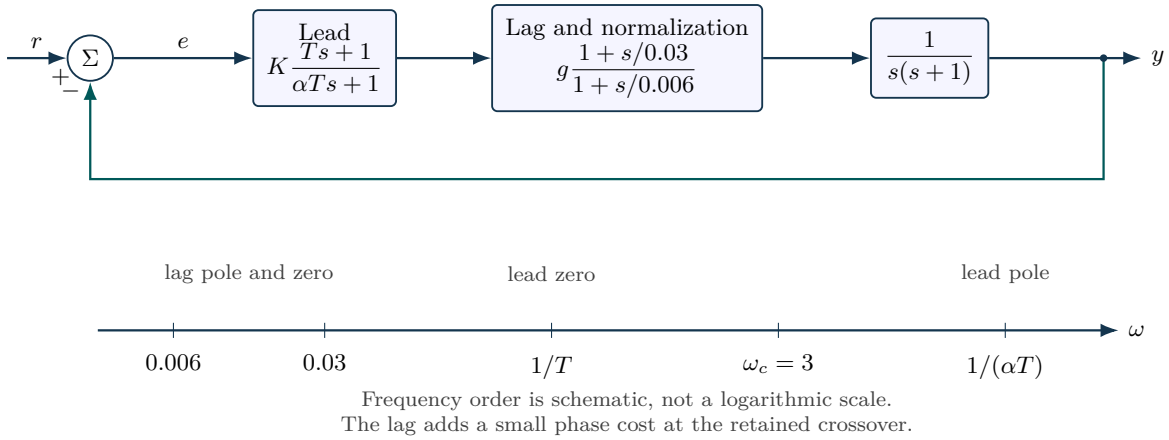
Worked reasoning. The plant magnitude at 3 is $1/(3\sqrt{10})$. At its phase peak the lead ratio has magnitude $1/\sqrt{\alpha}$. Hence $K(1/\sqrt{\alpha})/(3\sqrt{10}) = 1$, giving $K = 3\sqrt{10\alpha}$. With the preceding phase target this is approximately 4.7750. Using the plant-only gain $3\sqrt{10}$ would multiply loop magnitude by the additional lead factor and move the crossover.

(c) An implementation accidentally exchanges the calculated lead pole and zero. Determine the sign of its maximum phase contribution and explain why retuning scalar gain cannot repair that phase error.

Worked reasoning. The intended section has $0 < \alpha < 1$, with zero at $-1/T$ and farther-left pole at $-1/(\alpha T)$. Interchanging them forms the reciprocal phase response: its contribution is negative where the lead contribution was positive. Scalar renormalization can restore unity magnitude at 3 rad/s, but contributes no positive phase. The intended $+36.56505^\circ$ contribution becomes negative, so the phase specification is not recovered by gain adjustment.

3. Improving ramp tracking without silently relaxing phase specifications

Add a lag network to improve low-frequency tracking while preserving crossover. Carry the gain normalization through all steady-error calculations.



Frequency order is schematic, not a logarithmic scale.
The lag adds a small phase cost at the retained crossover.

Lead and lag sections are drawn separately so their gain normalization and frequency ordering can be analyzed independently.

(a) For the stable lead-compensated loop, calculate its velocity constant and unit-ramp error. State why the DC gain of G itself cannot be inserted as a finite number.

Worked reasoning. The loop has one integrator, and $K_v = \lim_{s\to 0} sC_{lead}G = C_{lead}(0) = K$. Thus $e_{ramp} = 1/K \approx 0.20942$. The plant has infinite DC gain because of its pole at zero, but the limit defining K_v cancels that integrator and remains finite. The unit-step error is zero under nominal stable unity feedback. These steady errors do not specify transient overshoot.

(b) Add $F(s) = (1 + s/0.03)/(1 + s/0.006)$ and multiply the controller by g so crossover remains at 3. Derive g, the new velocity constant and the new unit-ramp error.

Worked reasoning. At 3 rad/s, $|F| = \sqrt{1+100^2}/\sqrt{1+500^2}$. Choose $g = 1/|F| = \sqrt{250001/10001} \approx 4.99976$. Because $F(0) = 1$, the new velocity constant is gK, about 23.8739, and ramp error is $1/(gK)$, about 0.04189. Most of the low-frequency improvement comes from the renormalizing scalar gain. The explicit normalization convention must accompany any claim of a fivefold improvement.

(c) Require unit-ramp error no greater than 0.05 and phase margin at least 55 degrees. Using the resulting design, check both requirements together rather than rounding the margin.

Worked reasoning. The normalized lead–lag design has $K_v \simeq 23.8739$ and ramp error about 0.041887, satisfying the 0.05 requirement. Its lag phase is negative, so the margin is approximately 54.54165°, below 55 degrees. Thus this exact design fails the combined specification even though tracking improves. Rounding the margin to 55 hides a violation; the lead phase, crossover or lag placement must be redesigned.

4. Spending the remaining phase reserve on a real communication link

Evaluate the final lead–lag design after a pure communication delay is introduced. Keep strict stability and a desired robustness margin as separate requirements.

(a) Compute the lag section phase at the retained crossover and the resulting phase margin. Explain why a lag network can improve low-frequency tracking while slightly reducing margin.

Worked reasoning. At 3 rad/s the phase is $\arctan(100) - \arctan(500) \approx -0.45835°$. Since the scalar renormalization has zero phase and crossover is retained, the new margin is $55 - 0.45835 \approx 54.54165°$. The lag pole is closer to the origin than its zero, creating a band of negative phase. Placing both frequencies well below crossover limits that phase cost while permitting low-frequency gain improvement.

(b) Assume the final lead-lag loop has the single crossover 3 rad/s and phase margin from Problem 4(a). Calculate its first constant-delay boundary and the delay allowance if at least 40 degrees of margin must remain. State the assumptions.

Worked reasoning. A pure delay preserves magnitude and removes $3T_d$ radians at crossover. The first boundary is $T_{crit} = (54.54165\pi/180)/3 \approx 0.31731$ s. Retaining 40 degrees requires $T_d \leq [(54.54165 - 40)\pi/180]/3 \approx 0.08460$ s. This uses the nominal stable loop, one relevant crossover and a constant pure delay. Sensor dynamics, uncertain plant gain or time-varying delays require re-evaluation.

(c) A proposed link adds 0.09 s of constant pure delay. Does the model retain 40 degrees of margin? Is failure of that requirement equivalent to instability under the stated single-crossover assumptions?

Worked reasoning. The remaining margin is $54.54165° - 3(0.09)180°/\pi \simeq 39.07179°$, so the 40-degree requirement fails. However, 0.09 s is below the nominal first delay boundary of about 0.31731 s. Under the stated single-crossover pure-delay assumptions, the model remains stable while failing the stronger margin requirement. Neither conclusion covers arbitrary unmodeled poles, variable delay or actuator constraints.

5. A stable compensated output can still demand noisy actuator effort

Check algebraic stability and then trace measurement noise to both physical output and controller output. These are different performance channels.

(a) Derive the lead-only characteristic polynomial and use the cubic Routh inequality to check stability with the design formulas. Do not infer stability solely from an intended phase target.

Worked reasoning. From $1 + C_{lead}G = 0$, the polynomial is $\alpha T s^3 + (1 + \alpha T)s^2 + (1 + KT)s + K$. All coefficients are positive. The remaining condition is $(1 + \alpha T)(1 + KT) > \alpha TK$. Expanding the left side minus the right gives $1 + \alpha T + KT + \alpha KT^2 - \alpha TK$, which can be evaluated with the actual designed parameters and is positive. The numerical root check in the validation script independently verifies negative real parts; arbitrary positive K,T,alpha should not be substituted without checking.

(b) Let measurement be $y+n$ and feedback error be $r-y-n$. Derive the noise-to-output and noise-to-control transfers. Explain a high-frequency consequence of lead compensation.

Worked reasoning. Solving $y = GC(r - y - n)$ gives $Y/N = -CG/(1 + CG) = -T_c$, where T_c is complementary sensitivity. Control satisfies $U/N = -C/(1+CG) = -CS$. At high frequency, plant gain tends to zero and $S \to 1$, so control noise gain approaches $C(\infty)$. Lead-only $C(\infty) = K/\alpha$, larger than its DC gain K. Improved phase and bandwidth can therefore increase high-frequency actuator activity even when output noise remains small.

(c) For the final controller $C = gK(Ts+1)(1+s/0.03)/[(\alpha Ts+1)(1+s/0.006)]$, find its high-frequency gain and the corresponding limiting noise-to-control and noise-to-output gains.

Worked reasoning. The lead section tends to K/α and the lag section tends to $0.006/0.03 = 1/5$, so $C_\infty = gK/(5\alpha)$. Since $G(s) \sim 1/s^2$, the loop tends to zero. Therefore $U/N = -C/(1 + CG) \to -C_\infty$, whereas $Y/N = -CG/(1 + CG) \to 0$. A small high-frequency output response does not imply small actuator noise. This is an ideal high-frequency limit; a physical drive adds bandwidth and rate constraints.